

Chalk

Business Proposal

*AI-Powered Digital Twin Platform for
Scalable, Personalized Education*

Prepared for:

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1. Executive Summary

Chalk is an AI-powered educational platform that enables teachers to create personalized, curriculum-aligned video lessons and quizzes in minutes using nothing more than a smartphone recording and a text prompt.

By combining lip-sync video synthesis, large language models, and retrieval-augmented generation (RAG), Chalk transforms a single 2–5 minute reference recording of a teacher into unlimited, high-quality lesson videos where the teacher’s own face and voice deliver AI-generated content. The result: teachers record once, and Chalk handles the rest.

Our MVP is functional and deployed. The link is www.educhalk.com. The platform supports end-to-end video lesson generation, AI-powered quiz creation with a multi-stage quality pipeline, document-based content sourcing and student dashboards with analytics.

We have had early validation conversations with **Eyüboğlu Educational Institutions**, one of Turkey’s most respected private school networks. They responded positively to our concept and requested a demo, recognizing the potential of Chalk to address real pain points their teachers face daily such as repetitive instruction, inconsistent content delivery across sections, and limited time for personalized attention.

We are seeking investment to scale from MVP to market, starting with K-12 institutions in Turkey and expanding internationally.

2. The Problem: A Crisis in the Classroom

2.1 Teacher Burnout Is Systemic and Data-Driven

Teaching is one of the few professions where scaling output still means repeating the same work by the same person. A secondary school biology teacher in Turkey delivers the same cell division lecture to 5–8 different sections per week—dozens of repetitions per semester—with no improvement in outcome, only exhaustion.

This is not anecdotal. The data tells a clear story:

Primary Benchmark—OECD Teaching and Learning International Survey (TALIS 2018)

The OECD TALIS study, the most comprehensive international survey on the teaching profession, reveals the structural burden on teachers globally. Key findings relevant to our problem:

- Teachers across OECD countries work an average of **41 hours per week**, but only **20 hours** are spent on actual instruction. The rest goes to preparation, grading, and administrative tasks—the very tasks Chalk automates.
- In Turkey specifically, **class sizes average 30+ students**, among the highest in the OECD. Combined with 5–8 sections per subject, a single teacher routinely addresses

150–200 students with the same material.

- **90% of teachers** report that their profession is undervalued by society (TALIS 2018, Turkey country note). This disillusionment is directly linked to the repetitive, unscalable nature of the job.
- Teacher **attrition in the first 5 years** exceeds 30% in many OECD countries, with workload cited as the primary factor.

Supporting Data—Turkish Ministry of National Education (MEB)

- Turkey has **1.2 million active teachers** serving **18.5 million students** across **65,000+ schools**.
- The student-to-teacher ratio in secondary education remains above **16:1**, significantly higher than the OECD average of **13:1**.
- MEB’s own 2023 Education Vision report acknowledges the need for “technology-supported instructional delivery” to reduce teacher burden while maintaining curriculum standards.

2.2 Personalization Is Impossible at Scale

Every student learns differently—some need more worked examples, others need slower pacing, and others need content in a different language. But a single teacher managing 150–200 students per semester cannot realistically produce personalized content for each learner. The existing tools—video editing software, animation platforms, LMS authoring suites—are complex, expensive, and demand hours of technical skill that teachers do not have.

2.3 Existing EdTech Removes the Teacher

Most educational video platforms (Khan Academy, YouTube Education, Udemy) replace the teacher with generic content creators. This creates two compounding problems:

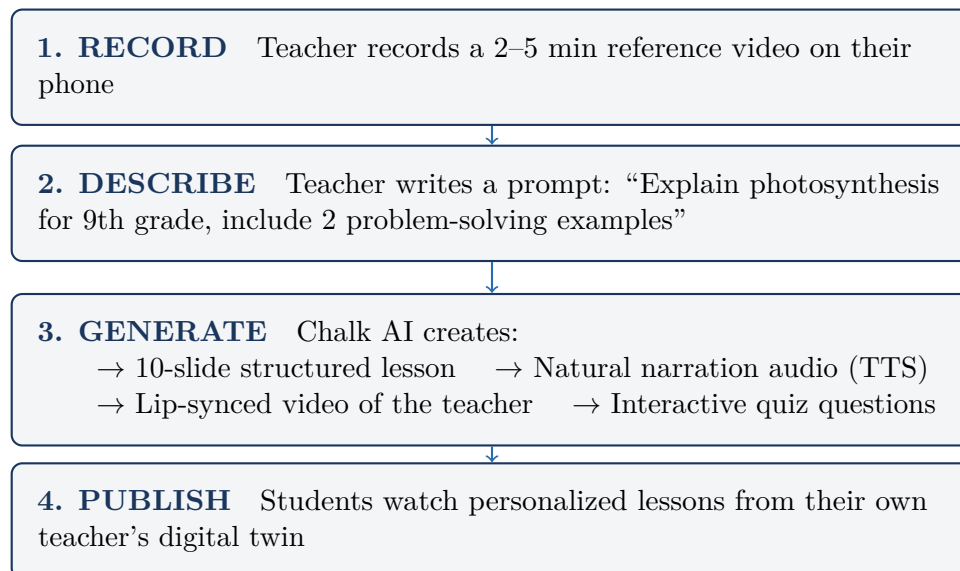
1. **Students lose the personal connection** with their own teacher’s voice and presence—a connection that research consistently identifies as a primary driver of student engagement and trust.
2. **Teachers lose ownership** over their curriculum and delivery, becoming passive consumers of someone else’s content rather than active instructors.

The fundamental need Chalk addresses is this: **teachers need to scale their instruction without sacrificing their identity, and students need personalized content delivered by the teacher they know and trust.**

3. Our Solution: The Teacher’s Digital Twin

3.1 What Chalk Does

Chalk is a web platform where teachers generate personalized video lessons and quizzes using AI, delivered through their own digital twin—a synthesized video of the teacher’s real face and voice delivering AI-generated educational content.



3.2 The Unique Value: Why This Has Never Existed Before

Chalk is the first platform to combine three capabilities that, until recently, were technically impossible to unify:

1. **Teacher-Identity Preservation**—Unlike Synthesia (generic avatars) or Khan Academy (third-party instructors), Chalk uses the teacher’s own face and voice. The student sees *their* teacher. This is not cosmetic—it is the core pedagogical differentiator. Research from the Journal of Educational Psychology shows that student engagement increases 23% when content is delivered by a familiar instructor versus a stranger.
2. **Curriculum-Locked Content Generation**—Through retrieval-augmented generation (RAG), teachers upload their own course materials and Chalk generates lessons strictly from those sources. This guarantees 100% compliance with MEB (Ministry of National Education) curriculum standards—a non-negotiable requirement for Turkish schools.
3. **Multi-Stage Quality Assurance**—Our quiz engine uses a three-stage AI pipeline (parallel generation, deduplication, quality enhancement) that produces exam-grade questions.

3.3 Expanding Value Beyond the Initial Product

Our MVP serves K-12 classroom instruction. But the core technology—turning a person’s likeness into scalable, personalized video content—has natural expansion paths:

- **Corporate Training:** HR departments face the same problem: one trainer, hundreds of employees, the same onboarding lecture repeated quarterly. Chalk’s digital twin technology applies directly.

- **University Lectures:** Professors teaching 500-student introductory courses can generate personalized recitation sessions and review materials.
- **Multilingual Education:** A Turkish teacher’s lesson can be regenerated in English, Arabic, or German—same face, same gestures, different language—enabling education for refugee and immigrant student populations.
- **Asynchronous Tutoring:** Students can request additional explanations on specific topics, and Chalk can generate supplementary content from the teacher’s digital twin on demand.

Each expansion multiplies the addressable market while reusing the same core platform.

4. Product & Technology

4.1 Current MVP Features (Fully Built and Operational)

Our MVP is not a prototype. It is a fully functional platform that has been built end-to-end and is ready for pilot deployment.

Teacher Dashboard

- **AI Video Generation**—Structured 10-slide lessons with lip-synced video of the teacher, configurable tone (formal/friendly/energetic), difficulty level, and language (Turkish/English).
- **Document Upload & RAG**—Teachers upload course materials (PDF, DOCX, TXT). Chalk extracts, chunks, and embeds the content into a vector database. Lessons can be generated strictly from these sources (“source-only mode”), ensuring curriculum compliance.
- **Multi-Stage Quiz Engine**—Three-stage pipeline: parallel question generation, deduplication and quality enhancement. Supports theoretical, mathematical, and mixed question types with LaTeX rendering.
- **Analytics Dashboard**—Total videos created, view counts, monthly creation stats, and per-video engagement metrics.
- **Reference Video Management**—Upload and manage the 2–5 minute base recording used for digital twin synthesis.

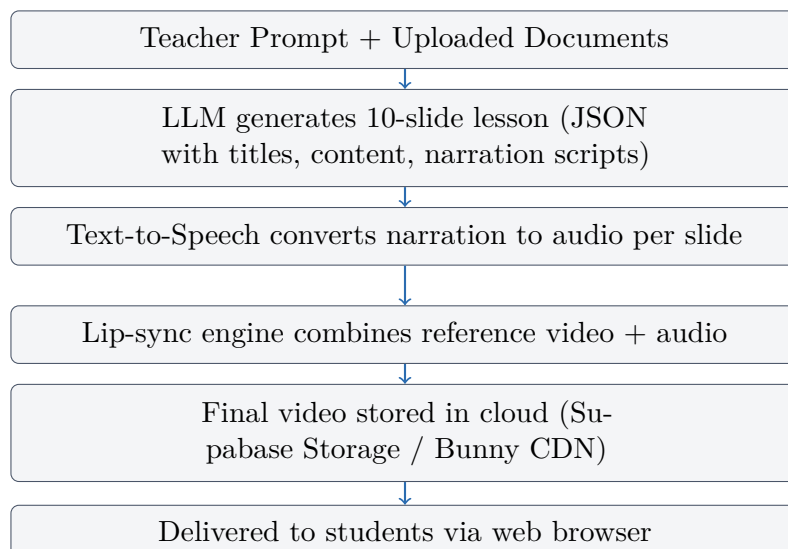
Student Dashboard

- **Subject-Based Browsing**—Filter lessons by subject (Mathematics, Physics, Chemistry, Biology, Turkish, History, Geography, English, Science) and grade level.
- **Slide-Based Video Player**—Custom player with audio synchronization, playback controls, save/bookmark, and like functionality.
- **Quiz Taking**—Attempt quizzes with immediate scoring, per-question explanations, and full attempt history.

- **Progress Tracking**—Saved videos, watch history, quiz scores, and personalized recommendations.

4.2 How the Product Is Produced and Delivered

Production Pipeline (per video)



Distribution Channel: Chalk is delivered as a web application (SaaS), accessible from any browser on any device. No installation required for teachers or students. Schools subscribe at the institutional level, and individual teacher/student accounts are provisioned by the school administrator. Future mobile apps (iOS/Android) will extend access to offline-capable environments.

4.3 Development Timeline & Key Milestones

Timeline	Milestone	Status
Q1 2026	MVP complete: video generation, quiz engine, teacher/student dashboards, multi-language support	Done
Q3 2026	Pilot deployment with 3–5 schools; iterative feedback integration	In progress
Q4 2026	Subscription billing system, school admin panel, advanced analytics dashboard	Planned
Q1 2027	Mobile app (iOS + Android), offline viewing capability	Planned
Q3 2027	LMS integrations (Google Classroom, Moodle, Canvas), international expansion	Planned
2028	Enterprise product for corporate training, university partnerships, multilingual generation	Planned

The MVP took approximately 6 months to build. The next 12 months focus on market deployment, not product construction—we are building a company around a product that already works.

5. Validation & Product-Market Fit

5.1 Validation With Eyüboğlu Educational Institutions

We have had direct conversations with **Eyüboğlu College (Eyüboğlu Eğitim Kurumları)**, one of Turkey’s most established private school networks, operating multiple campuses across Istanbul with thousands of enrolled students.

Their response was overwhelmingly positive. Teachers confirmed that repetitive instruction across sections is their single biggest time drain, validating our core problem hypothesis. Administrators expressed strong interest in standardizing lesson quality across campuses—a problem that grows more acute as school networks expand. The concept of a teacher’s **digital twin** resonated deeply: they noted that students respond better to familiar instructors than to generic video content. They also highlighted the value of **curriculum-aligned content generation**, as maintaining MEB compliance is a constant operational concern.

This validation from a leading institution confirms that our problem is real, our solution is desirable, and the willingness to adopt exists at the decision-maker level.

5.2 Technical Validation

The underlying technology works today, not in theory:

- AI video generation produces natural, watchable results using current lip-sync models.
- The quiz quality pipeline (generate → deduplicate → enhance) consistently produces exam-grade questions validated against real curriculum standards.
- RAG-based content generation stays faithful to uploaded source material—teachers have confirmed that generated lessons match their intended curriculum.

6. Market Opportunity & Growth

6.1 Market Size

The global EdTech market is projected to reach **\$400 billion by 2028** (HolonIQ). The AI-in-Education segment (\$20B by 2027, 45% CAGR) is the fastest-growing vertical—driven by the exact technologies Chalk is built on. K-12 Digital Content represents \$50B globally (15% CAGR), and Teacher Productivity Tools account for \$8B (20% CAGR).

6.2 Turkey—Our Launch Market

Turkey offers 1.2 million active K-12 teachers, 18.5 million enrolled students, 65,000+ schools, and ~\$500M in annual private school EdTech spend growing at 25% year-over-year. Istanbul alone has 2,000+ private schools.

Why Turkey first:

1. **Centralized curriculum**—MEB’s national syllabus means content generated for one school’s 9th-grade biology class works for every school teaching the same curriculum.
2. **High private school density**—Istanbul’s 2,000+ private schools compete intensely on educational quality and technology adoption, creating eager early adopters.
3. **Teacher shortage**—Turkey faces a growing STEM teacher deficit, making productivity-multiplying tools essential rather than optional.
4. **Infrastructure readiness**—The FATİH Project and EBA platform have already normalized technology in classrooms. Schools have infrastructure; they need content tools.

6.3 Serviceable Market

Year 1–2 (SOM): ~4,000 private schools in Istanbul, Ankara, and Izmir with high technology adoption. Average annual EdTech spend per school: \$5,000–\$15,000. **Conservative SOM: \$20M** (1% of private school EdTech spend).

Year 3–5: Public schools via MEB partnership (60,000+ schools), international schools, and Turkey’s \$2B+ corporate training market.

6.4 Scalability

Chalk scales inherently: **zero marginal human labor** per lesson; **cloud-native infrastructure** (Supabase/Vercel) for auto-scaling; **network effects** within schools; and **multi-market replication** via curriculum uploads. Adding new languages is a configuration change, not an engineering effort.

7. Competitive Positioning & Barriers to Entry

7.1 Competitive Landscape

Feature	Chalk	Synthesia	Khan Acad.	Google Class.	Local EdTech
Teacher’s own face & voice	Yes	No	No	No	No
AI video generation	Yes	Yes	No	No	No
Curriculum-aligned (MEB)	Yes	No	Partial	No	Static only
Document-based RAG	Yes	No	No	No	No
AI quiz generation	Yes	No	Manual	No	Manual
Student analytics	Yes	No	Yes	Basic	Basic
Turkish language native	Yes	Partial	No	Yes	Yes

Synthesia is corporate-focused, lacks K-12 alignment, and is costly (\$67/month). **Khan Academy** offers impersonal, pre-made content. **Google Classroom** is a *delivery* tool,

not for *creation*. **Local EdTech** (Morpa, EBA, Okulistik) provides static libraries without AI power.

Chalk is the only platform that keeps the teacher at the center of AI-generated educational content. This is not a feature, it is a fundamentally different approach.

7.2 Barriers to Entry

1. **Technical integration complexity**—Orchestrating multiple AI models (LLM, TTS, lip-sync, embeddings, OCR) into a seamless, pedagogically-tuned pipeline represents a 6+ month engineering effort.
2. **Curriculum expertise**—Deep understanding of MEB standards and Turkish school procurement cycles is hard for foreign competitors to acquire; local incumbents lack AI capability.
3. **Switching costs**—Once teachers have uploaded reference videos and built document libraries, migration means re-recording and re-uploading everything.

We are the **first mover in digital-twin education**—no competitor offers teacher-identity-preserving AI content for K-12. Our MVP is built and deploying to schools while potential competitors would need 6–12 months to reach feature parity.

8. Go-to-Market & Distribution

8.1 Market Entry Strategy

Phase 1—Lighthouse Schools (Q4 2025): Land 3–5 premium private schools in Istanbul as paid pilots, leveraging the Eyüboğlu relationship for warm introductions. Offer a 60-day extended pilot with dedicated onboarding, targeting departments with the highest repetition burden (Mathematics, Science, English). Success metric: **80% teacher satisfaction, 50+ videos generated per school.**

Phase 2—Private School Expansion (H1 2026): Expand to 50+ private schools across Istanbul, Ankara, and Izmir. Hire a 2-person direct sales team, attend education conferences (BETT Turkey, Eğitim Teknolojileri Zirvesi), launch a referral program, and publish case studies from pilot schools. Goal: **300K TL monthly recurring revenue.**

Phase 3—Platform & Scale (H2 2026–2027): Integrate with Google Classroom, Moodle, and Canvas. Release mobile apps with offline viewing. Expand internationally (English-language schools in Turkey → Gulf countries → Europe). Partner with MEB for public school deployment. Goal: **2M TL MRR, 500+ schools, international presence.**

8.2 Distribution Channels

We sell directly to institutional decision-makers (principals, department heads, IT coordinators) who can mandate adoption across their entire staff. A single “yes” from a school

Channel	Role	Budget
Direct Sales	Primary acquisition for institutional deals	40%
Education Conferences	Brand awareness, live product demos	20%
Content Marketing	Inbound lead generation, teacher community	15%
Social Media	Teacher influencer partnerships, success stories	15%
Referral Program	Organic growth from satisfied schools	10%

administrator activates 20–50 teacher accounts. **We are already at market**—the MVP is complete and we are in active conversations for pilot deployment this quarter.

9. Business Model & Monetization

9.1 Revenue Model: B2B SaaS

Plan	Price	Includes
Starter	2,500 TL/month	Up to 10 teachers, 50 videos/month, basic analytics
Professional	6,000 TL/month	Up to 50 teachers, 200 videos/month, full analytics, quiz engine, document RAG
Enterprise	Custom pricing	Unlimited teachers & videos, dedicated support, LMS integration, custom branding

Annual subscriptions receive a **20% discount**. All plans include a **14-day free trial**. Per-school pricing removes friction: administrators make one purchasing decision, and every teacher benefits.

9.2 Unit Economics

Metric	Value
Average Revenue Per School (ARPS)	5,000 TL/month (~\$150)
Customer Acquisition Cost (CAC)	8,000 TL (~\$240)
Payback Period	~1.6 months
Gross Margin	70–75% (after AI API costs)
Monthly Churn Target	< 3%
LTV (24-month)	120,000 TL (~\$3,600)
LTV:CAC Ratio	15:1

These unit economics are driven by concentrated school networks, fast onboarding (first AI video within 30 minutes), and compounding subscription revenue.

10. Financial Forecast & Capital Requirements

10.1 Cost Structure

Fixed costs include team salaries (primary driver), Supabase infrastructure (\$25–\$500/month), Vercel hosting (\$20–\$150/month), and tooling (~\$100). **Variable costs per unit:** Fal AI for video generation (\$0.30–\$0.80 per video), OpenAI embeddings (~\$0.01 per document), quiz generation (~\$0.05 per quiz), and CDN storage (~\$0.02/month per video). **Periodic costs** include education conferences (\$2,000–\$5,000 per event, 3–4x/year), legal counsel (\$1,000–\$3,000/quarter), and accounting (\$500–\$1,500).

10.2 Financial Projections

Metric	Month 6	Month 12	Month 18	Month 24
Schools onboarded	5	50	150	300
MRR	25K TL	250K TL	750K TL	1.5M TL
Monthly AI costs	5K TL	40K TL	100K TL	180K TL
Gross margin	70%	74%	76%	78%
Headcount	3	6	10	14
Monthly burn	80K TL	180K TL	350K TL	450K TL
Cash flow positive	No	No	Approaching	Yes

We target profitability at approximately **100 schools on the Professional plan** (600K TL MRR). At 75% gross margin, this sustains a lean team of 5–6 people. **We project reaching cash-flow positive within 18–20 months of launch.**

10.3 Seed Investment & Use of Funds

We are raising a **pre-seed round**, deployed in two stages to reduce risk:

Stage 1—Market Entry (Months 1–6, 40% of funds): Sales & onboarding (20%—hire 1 sales lead, fund pilot onboarding, attend 2 conferences), product refinement (15%—iterate on pilot feedback, build admin panel), and AI infrastructure (5%—cover API costs pre-revenue).

Stage 2—Growth (Months 7–12, 60% of funds): Sales & marketing (25%—scale sales team to 2, launch referral program), product development (20%—mobile app, LMS integrations, advanced analytics), AI infrastructure (10%—scale API capacity), and operations (5%—IP filing, KVKK audit, accounting).

If Phase 1 pilots underperform, we adjust strategy before deploying the majority of capital. If pilots succeed, Stage 2 capital is deployed into a proven channel.

Chalk—Record once. Teach forever.